

An Anderson–Rubin Test of Sectoral-Composition Optimality at the Municipality Level

Specification, Identification, and the Reduced-Form Equivalence

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Abstract

This document is the technical reference for the Anderson–Rubin (AR) optimality test that is the inferential object of the paper. It (i) derives the directional-derivative condition that grounds the null $H_0 : \beta = \mathbf{0}$ in the envelope theorem; (ii) maps the shift-share identification framework of [Borusyak, Hull, and Jaravel \(2022\)](#) (BJS) onto our political-turnover instrument and discusses the levels-versus-changes choice formally; (iii) writes the instruments out from primitives, including the office-specific pre-election window and the two timing variants; (iv) shows that the AR test of $H_0 : \beta = \mathbf{0}$ is exactly the joint Wald test on the instruments in the reduced-form GDP equation, and that this equivalence is what makes the estimator of script 54 a weak-instrument-robust optimality test; (v) treats the volume-channel control formally; and (vi) collects the inference questions raised by spatial correlation in shift-share designs. The document is intentionally compact and self-contained: all primitive objects (firm-level owner counts, party-exposure weights, cross-office shifts, and instruments) are defined here.

Setup and notation

We index municipalities by $m \in \{1, \dots, M\}$ ($M = 5,570$), sectors by $j \in \{1, \dots, J\}$, years by $t \in \{2002, \dots, 2017\}$, offices by $\ell \in \{M, G, P\}$ (mayor, governor, president), and parties by p . Lower-case Latin letters f, g index firms. Let $s(m)$ denote the state of municipality m . Cross-office channels are indexed by $c \in \{M, M\cdot P, M\cdot G, M\cdot G\cdot P\}$.

For office ℓ and year t , let $e_\ell(t)$ denote the most recent election year for office ℓ on or before t . The pre-election window is

$$T_t^\ell \equiv [e_\ell(t) - 4, e_\ell(t) - 1] \cap [2002, 2017]. \quad (1)$$

Within an electoral term, T_t^ℓ is constant in t ; at cycle boundaries, $e_\ell(t)$ advances and T_t^ℓ updates. The 2003 governor/president cycle (whose pre-window is 1998–2001, fully outside the data) is dropped.

The endogenous object is the vector of *sector employment shares*

$$s_{jmt}^{\text{emp}} \equiv \frac{n_{jmt}}{n_{mt}}, \quad n_{mt} \equiv \sum_{j=1}^J n_{jmt}, \quad \sum_{j=1}^J s_{jmt}^{\text{emp}} = 1, \quad (2)$$

where the firm universe is RAIS (Brazil’s formal-employment registry). Whenever no ambiguity arises we abbreviate $s_{jmt} \equiv s_{jmt}^{\text{emp}}$; the superscript “emp” is preserved when contrasting against the BNDES credit share s_{jmt}^B in (3). The decision to fix the firm universe at RAIS, taken in D24 (2026-05-06) and operationalized by the firm-support hybrid memo (D25, 2026-05-12), restricts the analyzed channel to *formal-sector composition*: politically driven shifts in non-formal employment (informal, MEI-unfiled, RAIS-Negativa-only) are by construction outside s_{jmt}^{emp} . Stacked across j , $s_{mt} \equiv (s_{1mt}, \dots, s_{Jmt})' \in \Delta^{J-1}$ lies on the unit simplex; $\iota' s_{mt} = 1$ and $\iota' \Delta s_{mt} = 0$ for any change. The outcome is municipal log real GDP, $Y_{mt} \equiv \log(\text{GDP}_{mt})$, and $\Delta Y_{mt} \equiv Y_{mt} - Y_{m,t-1}$.

Skeleton construction. The panel underlying (2) is a *contemporaneous unbalanced* skeleton: a cell (j, m, t) enters the panel if at least one RAIS firm with sector j is present in muni m in year t . Two operational details follow from the Phase 0 audits (A0.1–A0.3) of the firm-support hybrid plan:

1. *Zero-employee firms.* The audit (A0.2) finds the within-RAIS zero-employee rate at 0.0000%: the reconstructed RAIS panel does not retain *Negativa*-equivalent rows, so the contemporaneous-skeleton-includes-zero-employee-firms provision of the strategy memo is moot under the current extract and the contemporaneous skeleton is equivalent to a $n_{f,t} \geq 1$ requirement. The $n_{mt} = 0$ drop count is 121 muni-years across 36 munis (0.14%), with no muni zero across all 16 years.
2. *RAIS coverage bound.* Owner-CNPJs absent from the reconstructed RAIS panel constitute 7.64% of admin-observed firm-years (A0.1); this is the upper bound on the *Negativa*-recoverable firm mass (not a tight equality, since some Owner-only firms are non-RAIS-universe, e.g. informal/MEI). The corresponding employment-mass impact, computed by reweighting Owner-only firms to the contemporaneous-RAIS employment scale (A0.5), is bounded at 1.83% of municipal employment.

The skeleton admits two sensitivity variants used throughout robustness ([subsection 9.1](#)): *frozen*, which restricts the cell to firms active in $\tau \in T_t^M$, and *balanced*, which additionally requires presence in every post-election year of the cycle.

We propose the following notation for BNDES sector credit shares to keep them distinct from employment shares:

$$s_{jmt}^B \equiv \frac{\text{BNDES}_{jmt}}{\sum_k \text{BNDES}_{kmt}}, \quad (3)$$

where BNDES_{jmt} denotes BNDES disbursements to sector j in municipality m in year t . BNDES shares are the upstream mechanism through which the political shock is transmitted to employment composition.

The total-volume control is the unit-free ratio

$$\text{Vol}_{mt} \equiv \frac{\sum_j \text{BNDES}_{jmt}}{\text{GDP}_{m,t_0}}, \quad (4)$$

with t_0 the initial period of the panel ($t_0 = 2002$).

1 The optimality test: theory

Environment. A municipal economy in (m, t) allocates fixed factors across J sectors with the vector of shares $s_{mt} = (s_{1mt}, \dots, s_{Jmt})' \in \Delta^{J-1}$. Local output (or any local welfare proxy) is

$$Y_{mt} = Y(s_{mt}; \theta_{mt}),$$

where θ_{mt} collects municipality-time fundamentals unaffected by an infinitesimal reallocation of sector shares (e.g., pre-existing productivity, prices, and infrastructure). We require $Y(\cdot; \theta)$ to be twice continuously differentiable in s on the relative interior of Δ^{J-1} .

The directional-derivative condition. Let $\iota \equiv (1, \dots, 1)' \in \mathbb{R}^J$ denote the all-ones vector. A feasible reallocation direction is any $v \in \mathbb{R}^J$ with $\iota'v = 0$, so the perturbation $s_{mt} + \varepsilon v$ remains on the simplex for small ε . At an interior optimum $s_{mt}^* \in \text{int}(\Delta^{J-1})$ ¹ of $Y(\cdot; \theta_{mt})$, the first-order condition is

$$\left. \frac{d}{d\varepsilon} Y(s_{mt}^* + \varepsilon v) \right|_{\varepsilon=0} = \sum_{j=1}^J \frac{\partial Y(s_{mt}^*)}{\partial s_j} v_j = \nabla_s Y(s_{mt}^*) \cdot v = 0 \quad \text{for every } v \perp \iota. \quad (5)$$

Equivalently, the gradient is parallel to ι , the vector normal (orthogonal) to all feasible reallocation directions, $\nabla_s Y \propto \iota$, i.e. the social marginal returns to factors are equalized across sectors. This is the local-optimality benchmark.

Remark 1 (Envelope intuition). Equation (5) is the envelope condition for the planner's problem $\max_{s \in \Delta^{J-1}} Y(s; \theta)$: at an interior maximizer, infinitesimal feasible perturbations have zero first-order welfare effect ([Samuelson, 1947](#)). Rejection of (5) therefore identifies that the observed allocation is not at an interior maximum, and the sign pattern of the gradient prescribes the reallocation direction that would raise Y .

¹Here $\text{int}(\Delta^{J-1})$ denotes the relative interior of the simplex: all sector shares are strictly positive. Cells with zero employment in one or more sectors are boundary points, where the analogous optimality condition is a Kuhn-Tucker inequality rather than the equality in (5).

From the gradient to a regression. Let $\Delta s_{mt} \equiv s_{mt} - s_{m,t-1}$, with $\iota' \Delta s_{mt} = 0$. Since $Y(\cdot; \theta_{m,t-1})$ is twice continuously differentiable, Taylor's theorem (first-order Taylor expansion) around $s_{m,t-1}$ gives

$$Y(s_{mt}; \theta_{m,t-1}) - Y(s_{m,t-1}; \theta_{m,t-1}) = \nabla_s Y(s_{m,t-1}; \theta_{m,t-1})' \Delta s_{mt} + R_{mt}, \quad R_{mt} = O(\|\Delta s_{mt}\|^2). \quad (6)$$

Observed GDP changes also include changes in fundamentals between $t-1$ and t . Adding and subtracting $Y(s_{mt}; \theta_{m,t-1})$ gives the exact decomposition

$$Y(s_{mt}; \theta_{mt}) - Y(s_{m,t-1}; \theta_{m,t-1}) = \Delta Y_{mt} = \underbrace{Y(s_{mt}; \theta_{m,t-1}) - Y(s_{m,t-1}; \theta_{m,t-1})}_{\text{share-composition change at fixed fundamentals}} + \underbrace{Y(s_{mt}; \theta_{mt}) - Y(s_{mt}; \theta_{m,t-1})}_{\text{fundamental change at the new shares}}.$$

Substituting the Taylor expansion for the first bracket yields

$$\Delta Y_{mt} = \nabla_s Y(s_{m,t-1}; \theta_{m,t-1})' \Delta s_{mt} + R_{mt} + D_{mt}, \quad D_{mt} \equiv Y(s_{mt}; \theta_{mt}) - Y(s_{mt}; \theta_{m,t-1}). \quad (7)$$

Under local optimality, the first-order composition term is zero, but observed GDP changes need not be zero: they may still reflect changes in fundamentals D_{mt} and the approximation error R_{mt} .

Thus the empirical test is not whether ΔY_{mt} is zero. It is whether the component of Δs_{mt} induced by the instruments predicts ΔY_{mt} after conditioning on controls. This is valid only if the instruments are orthogonal to the residual components containing D_{mt} , R_{mt} , gradient heterogeneity, and measurement error.

Equation (7) and the logic described above directly motivate a first-difference equation,

$$\Delta Y_{mt} = \beta' \Delta s_{mt} + \rho' \Delta X_{mt} + e_{mt}^\Delta, \quad (8)$$

where β is interpreted as a common projected composition gradient in the within-simplex subspace. The residual e_{mt}^Δ contains unabsorbed fundamental changes D_{mt} , the Taylor remainder R_{mt} , heterogeneity in the local gradient, and measurement error. The vector X_{mt} collects the relevant controls.

The projected-gradient interpretation in levels. The directional-derivative condition (5) is a pointwise statement: at an interior optimum, the gradient is parallel to ι at that point. The decomposition (7) bridges from the pointwise condition to a regression in ΔY via Taylor expansion at the cell-specific $s_{m,t-1}$. A second bridge, expanding around a common reference allocation $\bar{s}$ in the relative interior of the simplex, delivers a regression in levels:

$$Y(s_{mt}; \theta_{mt}) = Y(\bar{s}; \theta_{mt}) + \nabla_s Y(\bar{s}; \theta_{mt})' (s_{mt} - \bar{s}) + \tilde{R}_{mt}, \quad \tilde{R}_{mt} = O(\|s_{mt} - \bar{s}\|^2). \quad (9)$$

Project the cell-level gradient $\nabla_s Y(\bar{s}; \theta_{mt})$ onto a common within-simplex gradient β and rearrange:

$$\begin{aligned} Y_{mt} = & \beta' s_{mt} + \underbrace{[Y(\bar{s}; \theta_{mt}) - \beta' \bar{s}]}_{\text{(A) cell-specific level}} \\ & + \underbrace{(\nabla_s Y(\bar{s}; \theta_{mt}) - \beta)' (s_{mt} - \bar{s})}_{\text{(B) gradient heterogeneity}} + \underbrace{\tilde{R}_{mt}}_{\text{(C) Taylor remainder}}. \end{aligned} \quad (10)$$

Absorbing (A)–(C) and measurement error into a single residual e_{mt}^L yields the levels analogue of (8),

$$Y_{mt} = \beta' s_{mt} + \rho' X_{mt} + e_{mt}^L, \quad (11)$$

where β is the same within-simplex projected composition gradient as in (8) and X_{mt} collects the relevant controls.

Equations (8) and (11) test the same null using different residual structures. None is a-priori small in the raw cross-section. Both nonetheless deliver the same null,

$$H_0 : \beta = \mathbf{0}, \quad (12)$$

the empirical counterpart of the directional-derivative condition (5): at an interior local optimum, the cell-level gradient lies in the span of ι , its projection onto the within-simplex subspace $\{v : \iota' v = 0\}$ is zero, and with the simplex normalization below the corresponding β vanishes.

Remark 2 (Why the Taylor scale is smaller than it looks). (10) appears to require the linearization to hold over the full cross-sectional spread of Brazilian sectoral mixes, where $\|s_{mt} - \bar{s}\|^2$ is large. It does not.

Municipality fixed effects absorb each muni’s own average composition. By the time the instrument acts, the cross-Brazilian dispersion is gone. The shift-share instrument then moves s_{mt} by only a few percentage points. The linear approximation only needs to hold locally—over the small neighborhood the instrument explores within each municipality. This is the same demand the differences form makes.

Two interpretive consequences. (i) When cell-level gradients are heterogeneous, $\hat{\beta}$ is an Imbens–Angrist-style weighted average of $\nabla_s Y(\bar{s}; \theta_{mt})$, weighted by IV-induced variation in each cell. $H_0 : \beta = \mathbf{0}$ therefore tests “the IV-weighted average gradient is zero,” not a literal pointwise optimum at every (m, t) . (ii) The differences and levels specifications use different weights and different evaluation points. Agreement between them is robustness; disagreement locates gradient heterogeneity across the simplex.

Simplex constraint. Because $\iota' s_{mt} = 1$ holds in every (m, t) cell, the regressor matrix of s_{mt} on a constant has reduced rank. Two operationally equivalent solutions:

1. Drop one sector j_0 and interpret each β_j as the effect of substituting one unit of employment share from j_0 to j .
2. Residualize s_{mt} on ι (within (m, t) this is degenerate) or, equivalently, project onto the within-simplex subspace.

We adopt option (1) throughout. All elements of the estimated $\hat{\beta}$ should be read as differences relative to j_0 , and the AR test on the remaining $J - 1$ coefficients is invariant to the choice of j_0 .

2 Identification strategy

We operationalize the levels equation (11) as a panel projection that absorbs the cell-specific level $Y(\bar{s}; \theta_{mt})$ (channel (A) of (10)) into municipality and year fixed effects—valid to the extent that this object is approximately additive in (m, t) as $\alpha_m + \delta_t$ —and partials out the level (volume) channel of the political shock with the volume control Vol_{mt} defined in (4). The resulting levels equation is

$$Y_{mt} = \alpha_m + \delta_t + \beta' s_{mt} + \lambda \text{Vol}_{mt} + \varepsilon_{mt}, \quad (13)$$

where β is the same within-simplex projected composition gradient as in (8) and (11), and ε_{mt} is the residual e_{mt}^L after the FE-and-volume partialling. The volume term enters because the same political-turnover shock that perturbs the sectoral mix also perturbs the aggregate level of credit; β identifies the composition channel only after the volume channel has been partialled out. Identification of β further requires the instrument to be orthogonal to the residuals—a condition formalized through the BJS framework below. Section 6 develops the volume control; section 7 treats the remaining controls.

Remark 3 (Why a level regression rather than a difference). We adopt (13) as the primary specification because each tier’s alignment Align_{mpt}^ℓ is generally constant within an electoral cycle for most municipalities and turns over at the inauguration year (mid-cycle changes through impeachment-style processes are too rare to drive identification). The first-difference instrument $\Delta \text{Align}_{mpt}^\ell$ is therefore a one-period pulse at $t = e_\ell + 1$ and is mechanically zero in years $\tau \in \{2, 3, 4\}$ of the cycle. This risks missing the bulk of the response as the shock can produce effect any time within the term, not just at inauguration.² The levels instrument Align_{mpt}^ℓ keeps the shock active for the entire cycle, so the shift-share can predict s_{mt} in any of the four post-election years in which the response materializes; the differences instrument throws away, by construction, the variation in years $\tau \in \{2, 3, 4\}$.

²Another reason to prefer the levels specification is that the policy timing (BNDES allocation, firm hiring, and the resulting composition response) can unfold over the full term, not at inauguration alone.

2.1 Cross-office channel structure

The political-economy mechanism through which BNDES allocation responds to party alignment is *cross-office*—it bites when the same party (or its electoral coalition) holds power simultaneously at the mayoral office and at one or more higher tiers. The mayor controls the local interface with BNDES (procurement, BNDES Card take-up at municipal counters, signaling to firms about which projects will be supported), but the bulk of BNDES disbursement decisions are made at the regional or national level, where governors and the federal executive have leverage. A firm in municipality m aligned with party p realizes an effective political-credit advantage when both ends of the intermediation chain align:

- the mayor of m belongs to p (local gatekeeping); and
- either (i) the president belongs to p or to a formal coalition partner of p (national allocation favoring coalition-aligned municipalities), or (ii) the governor of $s(m)$ belongs to p or its coalition (state-level discretion over BNDES regional offices and over the state apparatus that complements federal credit programs).

A national-only or governor-only alignment that does not pass through a sympathetic mayor lacks the local intermediary required for sector-specific credit to land in m and is not expected to drive s_{mt} . We therefore restrict the instrument set to channels that *contain mayoral alignment as a factor*.

Operationally, define the cross-office shifts

$$\text{Align}_{mpt}^{\text{M} \cdot \text{P}} \equiv \text{Align}_{mpt}^{\text{M}} \cdot \text{Align}_{pt}^{\text{P}}, \quad (14)$$

$$\text{Align}_{mpt}^{\text{M} \cdot \text{G}} \equiv \text{Align}_{mpt}^{\text{M}} \cdot \text{Align}_{s(m),pt}^{\text{G}}, \quad (15)$$

$$\text{Align}_{mpt}^{\text{M} \cdot \text{G} \cdot \text{P}} \equiv \text{Align}_{mpt}^{\text{M}} \cdot \text{Align}_{s(m),pt}^{\text{G}} \cdot \text{Align}_{pt}^{\text{P}}, \quad (16)$$

where $\text{Align}_{pt}^{\text{P}} = 1$ if party p is in the federal presidential coalition (leading party or formal coalition partner) at time t , and $\text{Align}_{s(m),pt}^{\text{G}} = 1$ if p is in the gubernatorial coalition of state $s(m)$ at t . The coalition definition on the higher tiers reflects that BNDES allocation under any given mandate is shared across coalition members, not monopolized by the leading party. The pure $\text{Align}_{mpt}^{\text{M}}$ represents the standalone mayoral channel (capturing local-only political capital, e.g. municipal procurement and local-bank signaling).

Office-specificity of Brazilian coalitions. Brazilian electoral coalitions (*coligações*) are formed *per office per election*: the coalition supporting a presidential candidate is a distinct legal object from the coalition supporting a gubernatorial candidate in any given state, and from the coalition supporting a mayoral candidate in any given municipality. Constitutional Amendment 52/2006 (modifying Art. 17, §1) explicitly guarantees party autonomy “without obligatory linkage between candidacies at national, state, district or municipal level.” Two parties allied for a mayoral race in muni m may run on rival slates for the gubernatorial race in state $s(m)$ in the same election cycle, and either configuration may be unrelated to their national-presidential alignment. We therefore treat Align^{M} , Align^{G} , and Align^{P} as separately measured objects, each defined from the office-specific *electoral coalition* (the registered *coligação* on the candidate slate of office ℓ at election e_ℓ).

Before EC 52/2006, TSE Resolution 20.993/2002 imposed *verticalização* on the 2002 and 2006 elections, requiring state-level coalitions to align with the national presidential coalition. Within the 2002–2017 sample window, this constraint binds the 2002 and 2006 electoral cycles (gubernatorial inaugurations 2003 and 2007; post-election years 2003–2010).

2.2 Firm-level primitives

Here we define the firm-level primitives from which the sector-level exposures $w_{jmp,t}^\ell$ and firm-level instruments FA_{fmt}^c are built. The firm-level construction is the basis for the

Owner counts and firm support. Let $L_{f,p,t}$ denote the number of owners of firm f affiliated with party p at time t , and $L_{f,t}$ the total number of owners of firm f at time t (including owners not affiliated with any party), so

that $\sum_p L_{f,p,t} \leq L_{f,t}$. Let $\mathcal{F}(j, m)$ denote the firm support of cell (j, m) : the set of firms that ever appear in sector j , municipality m over the panel. The pre-period firm support associated with mayoral inauguration year e_M is

$$\mathcal{F}_{jm, e_M}^{\text{pre}} \equiv \{ f \in \mathcal{F}(j, m) : L_{f,s} > 0 \text{ for some } s \in T_{e_M}^M \},$$

i.e. the firms in (j, m) active at least one year of the mayoral pre-election window.

Firm-level baseline party exposure. The baseline exposure of firm f to party p for office ℓ at year t pools affiliated and total owner counts across the pre-election window:

$$\omega_{fp,t}^\ell \equiv \frac{\sum_{s \in T_t^\ell} L_{f,p,s}}{\sum_{s \in T_t^\ell} L_{f,s}}, \quad \sum_p \omega_{fp,t}^\ell \leq 1. \quad (17)$$

This pooled-count construction weights each owner-year observation equally rather than each calendar year equally: a year in which firm f has many owners contributes more to the baseline than a year in which it has few. Equivalently, $\omega_{fp,t}^\ell$ is the owner-year-weighted average of the annual affiliation shares $L_{f,p,s}/L_{f,s}$. The residual $1 - \sum_p \omega_{fp,t}^\ell$ is the unaffiliated-owner share at the firm level. The exposure is office-specific because the pre-election window T_t^ℓ is office-specific, but it does not vary across municipalities in the case of firms that operate in multiple municipalities.

Primary channel-specific baseline window. For the cross-office instruments, the primary exposure window is the pre-earliest-election window for the offices entering the channel. Let

$$\mathcal{O}(M) = \{M\}, \quad \mathcal{O}(M \cdot P) = \{M, P\}, \quad \mathcal{O}(M \cdot G) = \{M, G\}, \quad \mathcal{O}(M \cdot G \cdot P) = \{M, G, P\}.$$

Define

$$e_c^{\min}(t) \equiv \min_{\ell \in \mathcal{O}(c)} e_\ell(t), \quad T_t^c \equiv [e_c^{\min}(t) - 4, e_c^{\min}(t) - 1] \cap [2002, 2017]. \quad (18)$$

The channel-specific firm exposure is

$$\omega_{fp,t}^c \equiv \frac{\sum_{s \in T_t^c} L_{f,p,s}}{\sum_{s \in T_t^c} L_{f,s}}. \quad (19)$$

For the pure mayoral channel, $T_t^c = T_t^M$. For the cross-office channels, T_t^c is the last four-year window before the last election among the offices in the channel. The associated pre-period firm support is

$$\mathcal{F}_{jm,t}^{\text{pre},c} \equiv \{ f \in \mathcal{F}(j, m) : L_{f,s} > 0 \text{ for some } s \in T_t^c \}.$$

Firm-level cross-office instrument. For channel $c \in \{M, M \cdot P, M \cdot G, M \cdot G \cdot P\}$, the firm-level cross-office instrument is

$$\text{FA}_{fpt}^c \equiv \sum_p \omega_{fp,t}^c \text{Align}_{mpt}^c, \quad (20)$$

i.e. firm f 's pre-earliest-election baseline exposure for channel c , inner-producted with the channel- c shift in municipality m at time t .

Remark 4 (Primary exposure window). The primary specification uses the pre-earliest-election window T_t^c in (18). This choice is based on the predetermination rationale: the baseline exposure should be measured before any of the offices in the channel can turn over, so that it is not contaminated by post-treatment changes in owner-party affiliation. Since mayoral and gubernatorial/presidential elections alternate every two years, the pre-earliest window is mechanically older for part of each cycle and may be a noisier measure of the relevant affiliation stock. We therefore report a mayoral-window exposure, w_{jmt}^M used for all channels, as the mechanism-aligned robustness. A higher-tier pre-window is a second timing robustness for cases in which owner affiliation responds mainly to federal or state electoral incentives.

2.3 Sector-level exposure weights

Building the sector-level instruments from the firm-level exposures (17) requires two independent choices: across-sector normalization and within-cell aggregation. The former is the choice of the universe over which the sectoral shares are normalized; the latter is the choice of how to aggregate firm-level exposures within a cell. There are two natural normalization universes—within-cell (cell is its own universe) or within-municipality (municipality is the universe)—and three natural aggregation methods—owner-count, equal-firm, or employment weights. We adopt as primary configuration the within-municipality normalization and owner-count aggregation, and report the rest as robustness. The window choice was fixed in Remark 4. The rest of this subsection (i) defines the within-cell intensity weights, which are the cleanest notation for the BJS conditions and aggregation identity that follow; (ii) records that identity; (iii) defines Variants A and B.

Sector-level baseline exposure (within-cell intensity, robustness). The within-cell intensity-weighted baseline exposure aggregates firm-level exposures (17) across firms within (j, m) using owner-count weights, with the office- ℓ pre-election window:

$$w_{jmt}^\ell \equiv \frac{1}{|T_t^\ell|} \sum_{s \in T_t^\ell} \frac{\sum_{f \in \mathcal{F}(j,m)} L_{f,p,s}}{\sum_{f \in \mathcal{F}(j,m)} L_{f,s}}, \quad \sum_p w_{jmt}^\ell \leq 1. \quad (21)$$

The residual $1 - \sum_p w_{jmt}^\ell$ is the unaffiliated-owner share of cell (j, m) , which contributes zero to Z_{jmt}^ℓ but enters the exposure control $EC_{jmt}^\ell \equiv \sum_p w_{jmt}^\ell$ that absorbs total political connectedness of (j, m) .

For the primary cross-office instruments we use the same construction with the channel-specific pre-earliest window T_t^c : w_{jmt}^c is (21) with T_t^ℓ replaced by T_t^c . For the pure mayoral channel this is exactly w_{jmt}^M ; for the cross-office channels it is the exposure measured before the earliest election entering the current cross-office state. The mayoral-window robustness instead uses w_{jmt}^M for all channels.

Owner-count normalizer. Let

$$N_{jmt}^\ell \equiv \sum_{f \in \mathcal{F}(j,m)} L_{f,t}, \quad (22)$$

the total number of owners across firms in cell (j, m) at time t . The notation N_{jmt}^c denotes the same normalizer evaluated on the channel-specific support $\mathcal{F}_{jmt}^{\text{pre},c}$.

Frozen pre-election window. The primary window T_t^c in (18) is *frozen* relative to the post-election alignment shocks: it never overlaps the post-inauguration interval over which Align_{mpt}^c is treated as the exogenous shift. This satisfies the predetermined-shares logic of Goldsmith-Pinkham, Sorkin, and Swift (2020) and supplies the share-level predetermination that Borusyak, Hull, and Jaravel (2022) use to translate shock-level orthogonality (Condition 1) into unit-level orthogonality. The within-cell sum $EC_{jmt}^c = \sum_p w_{jmt}^c \in [0, 1]$ measures the fraction of cell (j, m) 's pre-window owner mass affiliated with any party; the residual slack $_{jmt}^c \equiv 1 - EC_{jmt}^c$ is unaffiliated mass that contributes zero to Z_{jmt}^c . In the language of Borusyak, Hull, and Jaravel (2022, §4.4), this is the *incomplete shares* setup, and the cell-level slack must be carried as a control in the muni panel. Operationally we form the per-cell slack column `slack_frozen_mt` on the frozen window and include it among the controls in the muni panel of (13)–(44); on/off perturbations are documented in subsection 9.1.

Sector exposure as owner-count-weighted firm average. Substituting (17) into (21) and rearranging gives the equivalent representation

$$w_{jmt}^\ell = \sum_{f \in \mathcal{F}(j,m)} \frac{L_{f,t}}{N_{jmt}^\ell} \omega_{fp,t}^\ell. \quad (23)$$

The sector-level baseline party exposure is the owner-count-weighted average of firm-level exposures across the firm support $\mathcal{F}(j, m)$.³ Equation (23) is the bridge between the firm-level instrument (20) and the sector-level instruments (29)–(32).

The intensity-based exposure $w_{jmp,t}^c$ in (21) normalizes owner-party counts *within* each (j, m) cell, so it captures the intensity of political affiliation in the cell but discards information about the cell's mass in the local economy. For predicting sector employment shares s_{jmt} this is a known weakness of the intensity-only shift-share (29)–(32): a small cell with a high aligned-fraction generates a large instrument value but can move little employment when credit arrives, while a large cell with a moderate aligned-fraction can move a lot. This subsection records two alternative exposure specifications, both predetermined on T_t^c and both preserving the shift-share decomposition, that bring mass back into the share component.

Pre-window employment primitives. Let $E_{f,t}$ denote firm f 's employment at time t . Define the pre-window firm employment, sector-muni pre-window employment, and muni pre-window employment as the natural averages over the channel-specific pre-earliest-election window T_t^c :

$$\bar{E}_{f,t}^c \equiv \frac{1}{|T_t^c|} \sum_{s \in T_t^c} E_{f,s}, \quad \bar{E}_{jm,t}^c \equiv \sum_{f \in \mathcal{F}(j,m)} \bar{E}_{f,t}^c, \quad \bar{E}_{m,t}^c \equiv \sum_{j'} \bar{E}_{j'm,t}^c. \quad (24)$$

The pre-window matches the one used for owner-party counts in (17) and (21), so the predetermination argument that justifies $w_{jmp,t}^c$ extends without modification to any object built from $\bar{E}$.

Variant A: muni-relative aligned-owner share (primary). The primary spec keeps the owner-count primitives but replaces the within-cell normalization in (21) with a muni-level normalization over affiliated owners. Let $\bar{L}_{jmp,t}^c \equiv \sum_{s \in T_t^c} \sum_{f \in \mathcal{F}(j,m)} L_{f,p,s}$ be the pre-window owner-years in (j, m) affiliated with party p , and $\bar{L}_{m,t}^{c,\text{affil}} \equiv \sum_{j'} \sum_{p'} \bar{L}_{j'mp',t}^c$ the muni-level pre-window total of owner-years affiliated with any party. Define

$$\tilde{w}_{jmp,t}^{c,\text{own}} \equiv \frac{\bar{L}_{jmp,t}^c}{\bar{L}_{m,t}^{c,\text{affil}}}, \quad \tilde{Z}_{jmt}^{c,\text{own}} \equiv \sum_p \tilde{w}_{jmp,t}^{c,\text{own}} \text{Align}_{mpt}^c. \quad (25)$$

The share $\tilde{w}_{jmp,t}^{c,\text{own}}$ is the fraction of the muni's politically-engaged pre-window owner-mass that sits in sector j at party p ; it carries information about cell size through $\bar{L}_{jmp,t}^c$ that the intensity $w_{jmp,t}^c$ discards. Both numerator and denominator are pre-window quantities, so $\tilde{w}_{jmp,t}^{c,\text{own}}$ is predetermined relative to the shift Align_{mpt}^c . Across sectors, $\sum_j \tilde{Z}_{jmt}^{c,\text{own}} \in [0, 1]$ equals the muni's share of pre-window affiliated owners whose party is aligned under channel c .

Variant B: muni-relative aligned-employment share (within-cell-aggregation robustness). This robustness keeps the muni-level normalization but swaps owner-mass for employment-mass at every aggregation level. Using the firm-level instrument (20), define

$$\tilde{Z}_{jmt}^{c,\text{emp}} \equiv \frac{\sum_{f \in \mathcal{F}(j,m)} \bar{E}_{f,t}^c \text{FA}_{fmt}^c}{\bar{E}_{m,t}^c}. \quad (26)$$

This is the pre-window employment of firms in (j, m) whose owners are aligned under channel c (with $\text{FA}_{fmt}^c \in [0, 1]$ as the per-firm alignment-intensity weight), expressed as a fraction of the muni's total pre-window employment. The same object admits the shift-share decomposition

$$\tilde{Z}_{jmt}^{c,\text{emp}} = \sum_p \tilde{w}_{jmp,t}^{c,\text{emp}} \text{Align}_{mpt}^c, \quad \tilde{w}_{jmp,t}^{c,\text{emp}} \equiv \frac{\sum_{f \in \mathcal{F}(j,m)} \bar{E}_{f,t}^c \omega_{fp,t}^c}{\bar{E}_{m,t}^c}. \quad (27)$$

³The simple (equal-firm-weight) average $\bar{\omega}_{jmp,t}^c \equiv |\mathcal{F}(j, m)|^{-1} \sum_f \omega_{fp,t}^c$ differs from (23) whenever firm-level exposure $\omega_{fp,t}^c$ covaries with firm size $L_{f,t}$. Owner-count weighting matches the empirical mass of political exposure in the cell and is the construction used throughout the pipeline.

The share admits a size-times-intensity decomposition

$$\tilde{w}_{jmp,t}^{c,emp} = \underbrace{\frac{\bar{E}_{jm,t}^c}{\bar{E}_{m,t}^c}}_{\text{sector size}} \underbrace{\bar{\omega}_{jmp,t}^{c,E}}_{\text{employment-weighted intensity}}, \quad \bar{\omega}_{jmp,t}^{c,E} \equiv \sum_{f \in \mathcal{F}(j,m)} \frac{\bar{E}_{f,t}^c}{\bar{E}_{jm,t}^c} \omega_{fp,t}^c, \quad (28)$$

in which $\bar{\omega}_{jmp,t}^{c,E}$ is the employment-weighted within-cell average of firm-level intensities — the direct analogue of (23) with owner-count weights replaced by employment weights. Both factors are pre-window and predetermined; the shift sits exclusively in Align_{mpt}^c as in the base spec.

Comparison of A and B. The variants address different facets of the mass-weighting concern. Variant A modifies the across-sector scaling: muni-level political mass is allocated to sectors in proportion to where the aligned *owners* are. Variant B modifies *both* the firm-to-sector aggregation (23) and the across-sector scaling, replacing owner-mass by *employment-mass* everywhere. The two coincide in within-muni ranking only when firm-level employment is proportional to owner counts; in industry-towns, where a small number of large employers carry most of the employment but a minority of the owners, Variant B will rank sectors differently from Variant A and from the base intensity spec. Variant B is preferred on the mechanism-fit argument that the conversion of credit to hires operates through firm size in employment, not in owners; Variant A is the conservative analogue that preserves the owner-count primitives already used throughout (17)–(23).

BJS conditions. Condition 1 and Condition 2 carry through for both variants under the same argument as (29)–(32): the shift Align_{mpt}^c is unchanged, and the share components are pre-window objects. Condition 3 is the condition where the variants can deviate from the base spec, because muni-relative normalization concentrates exposure on large-mass cells; the Herfindahl $\sum_j (\tilde{Z}_{jmt}^{c,\bullet})^2$ should be reported alongside the base-spec Herfindahl in robustness diagnostics, since industry-towns load disproportionately on a single sector under Variants A and B.

Exposure controls. The base-spec exposure control $\text{EC}_{jm,t}^c \equiv \sum_p w_{jmp,t}^c$ adapts mechanically: $\widetilde{\text{EC}}_{jm,t}^{\text{own}} \equiv \sum_p \tilde{w}_{jmp,t}^{c,\text{own}}$ is cell (j,m) 's share of muni affiliated owners under Variant A, and $\widetilde{\text{EC}}_{jm,t}^{\text{emp}} \equiv \sum_p \tilde{w}_{jmp,t}^{c,emp}$ its employment-mass analogue under Variant B. The muni-level political-mass scales $\log \bar{L}_{m,t}^{c,\text{affil}}$ (Variant A) and $\log \bar{E}_{m,t}^c$ (Variant B) are added to the controls in (13) to absorb the level of muni political mass without absorbing the across-sector tilt that identifies β .

Implementation. The primary instrument in first-stage and AR tables is $\tilde{Z}_{jmt}^{c,\text{own}}$ (Variant A). The intensity-based Z_{jmt}^c in (29)–(32) is reported as the across-sector-normalization robustness and $\tilde{Z}_{jmt}^{c,emp}$ (Variant B) as the within-cell-aggregation robustness. The three are non-nested but share the same identification skeleton; differences across them are informative about which mass concept predicts sector employment shares in the data.

2.4 The shift-share form of the cross-office instruments

We present the shift-share form using the within-cell intensity weights $w_{jmp,t}^c$ from (21); this is the cleanest notation for the BJS derivation that follows. The primary instrument $\tilde{Z}_{jmt}^{c,\text{own}}$ is the analogous shift-share with the muni-level weights of (25) substituted for $w_{jmp,t}^c$. With $w_{jmp,t}^c$ and the firm-level instrument (20) in hand, the four

sector-level levels instruments are

$$Z_{jmt}^M \equiv \sum_p w_{jmp,t}^M \text{Align}_{mpt}^M, \quad (29)$$

$$Z_{jmt}^{M \cdot P} \equiv \sum_p w_{jmp,t}^{M \cdot P} \text{Align}_{mpt}^{M \cdot P}, \quad (30)$$

$$Z_{jmt}^{M \cdot G} \equiv \sum_p w_{jmp,t}^{M \cdot G} \text{Align}_{mpt}^{M \cdot G}, \quad (31)$$

$$Z_{jmt}^{M \cdot G \cdot P} \equiv \sum_p w_{jmp,t}^{M \cdot G \cdot P} \text{Align}_{mpt}^{M \cdot G \cdot P}. \quad (32)$$

For backward compatibility with the rest of the document we let Z_{jmt}^c denote a generic channel $c \in \{M, M \cdot P, M \cdot G, M \cdot G \cdot P\}$ and Align_{mpt}^c its shift. The changes variant is

$$\Delta Z_{jmt}^c \equiv \sum_p w_{jmp,t}^c \Delta \text{Align}_{mpt}^c, \quad \Delta \text{Align}_{mpt}^c \equiv \text{Align}_{mpt}^c - \text{Align}_{mpt,t-1}^c, \quad (33)$$

which is non-zero in any year in which any of the offices entering c turns over.

The owner-count-weighted aggregation identity. Each Z_{jmt}^c is a within-region inner product of *shares* $w_{jmp,t}^c$ (sector-municipality exposure to party p , on the pre-earliest-election channel window) and *shifts* Align_{mpt}^c (whether p controls the offices defining c at time t). It is the sector-aggregated counterpart of the firm-level instrument (20): substituting (23) into (29)–(32) and reordering sums gives

$$Z_{jmt}^c = \sum_p w_{jmp,t}^c \text{Align}_{mpt}^c = \sum_p \left[\sum_{f \in \mathcal{F}_{jm,t}^{\text{pre},c}} \frac{L_{f,t}}{N_{jm,t}^c} \omega_{fp,t}^c \right] \text{Align}_{mpt}^c = \sum_{f \in \mathcal{F}_{jm,t}^{\text{pre},c}} \frac{L_{f,t}}{N_{jm,t}^c} \text{FA}_{fmt}^c. \quad (34)$$

The sector instrument is a weighted average of the firm-level instruments across firms in the pre-period support, with weights given by each firm's share of total owner-counts in the cell. The identity (34) verifies diagnostically: collapsing firm instruments by these weights recovers the sector instrument.

2.5 BJS conditions, mapped to our setting

Borusyak, Hull, and Jaravel (2022) treat shift-share IVs as IV estimators in a *shock-level* sample, where the unit of identification is a (shock) rather than an (m, t) cell. In our notation the “shocks” are office-party-municipality cells $k \equiv (\ell, p, m, t)$ with shifts $g_k \equiv \text{Align}_{mpt}^\ell$ (or $\Delta \text{Align}_{mpt}^\ell$) and exposures $w_{jmp,t}^\ell$. We restate their three primitive identification assumptions and map each to our setting.

Condition 1 (BJS-1: Quasi-random shock assignment). Conditional on a vector of shock-level controls q_k , $\mathbb{E}[g_k \mid \{w_{jmp,t}^\ell\}_j, q_k] = \mu(q_k)$ is the same for shocks with different exposure profiles. Equivalently, in our panel: among (ℓ, p, m, t) cells with the same controls q_k , who is aligned is as good as randomly assigned with respect to municipal GDP residuals.

Condition 2 (BJS-2: Many uncorrelated shocks). The number of effective shocks K_{eff} is large, and shocks are uncorrelated with each other after conditioning on q_k .

Condition 3 (BJS-3: Non-concentration of average exposure). The Herfindahl of the average exposure weights across shocks is small, so no single shock drives the IV estimate.

What each buys. **Condition 1** is the analogue of the exclusion restriction. It is what allows the instrument Z_{jmt}^ℓ to be exogenous in the structural GDP equation. **Condition 2** provides the law-of-large-numbers justification that shock-level orthogonality of g_k to municipal residuals translates into unit-level (muni-year) orthogonality of Z_{jmt}^ℓ to ε_{mt} . **Condition 3** ensures that the asymptotic distribution of the IV estimator is non-degenerate and not driven by a handful of large-exposure shocks.

What can violate Condition 1 in our setting. The political-alignment shock Align_{mpt}^ℓ is not literally random: parties win office because voters prefer them. Quasi-random assignment is plausible only after conditioning on a sufficient set q_k of shock-level controls. The relevant candidates for q_k are: (a) state-by-year fixed effects (so identification is within-state contests); (b) lagged municipal political controls (a party's pre-period vote share); (c) lagged GDP and employment trends (so a party that systematically wins in growing municipalities is not confounded with growth); and (d) sector-by-year fixed effects in the first stage (section 4) so that nationwide sectoral cycles are absorbed at the shock level. The empirical strategy includes muni and year fixed effects, exposure controls $\text{EC}_{jm,t}^\ell$, and (in robustness) a pre-period GDP trend; see section 7.

2.6 Levels versus changes: a formal sufficient condition

The same shift-share construction admits two timing variants of the shift, applied to each channel $c \in \{M, M \cdot P, M \cdot G, M \cdot G \cdot P\}$: $\text{Align}_{mpt}^c \in \{0, 1\}$ in levels, or $\Delta \text{Align}_{mpt}^c \in \{-1, 0, +1\}$ in changes. They differ in how the identifying variation is distributed over time. Let E_c denote the set of inauguration years for any office defining c : $E_M = \{2005, 2009, 2013, 2017\}$, while $E_{M \cdot P} = E_{M \cdot G} = E_{M \cdot G \cdot P} = \{2003, 2005, 2007, 2009, \dots, 2017\}$ is the union of mayoral and gubernatorial/presidential inaugurations.

Proposition 1 (Sufficient conditions for the levels and changes instruments). *Suppose $Y_{mt} = \alpha_m + \delta_t + \beta' s_{mt} + \lambda \text{Vol}_{mt} + \varepsilon_{mt}$ with $\mathbb{E}[\varepsilon_{mt} \mid \alpha_m, \delta_t] = 0$, and let $\mathcal{H}_{m,t-1} \equiv \sigma(Y_{m,t'}, \text{Vol}_{m,t'}, s_{m,t'}; t' < t)$ be the pre-period information set.*

(i) *The levels instrument Z_{jmt}^c satisfies the BJS quasi-random assignment condition Condition 1 if, between consecutive inauguration dates $e, e' \in E_c$ with $e < e'$,*

$$\mathbb{E}[\text{Align}_{mp,e+\tau}^c \mid \{w_{jmp,e}^c\}_j, \mathcal{H}_{m,e-1}, q_k] = \mu_\tau(q_k), \quad \tau = 1, \dots, e' - e, \quad (35)$$

i.e. the channel- c alignment status is uncorrelated with prior municipal trajectories conditional on q_k , throughout the entire post-inauguration interval.

(ii) *The changes instrument ΔZ_{jmt}^c satisfies Condition 1 if, for every $t \in E_c$,*

$$\mathbb{E}[\Delta \text{Align}_{mpt}^c \mid \{w_{jmp,t}^c\}_j, \mathcal{H}_{m,t-1}, q_k] = \mu(q_k), \quad (36)$$

i.e. each channel- c turnover is uncorrelated with prior trajectories conditional on q_k .

Discussion. Equation (35) requires conditional mean-independence of the channel- c alignment from pre-period outcomes at every horizon within the relevant inter-inauguration interval. Equation (36) requires it only at the moments of turnover. The latter is empirically weaker: turnover is closer to a discontinuity at each election date and need only be unrelated to the trajectory entering that election, not to the entire path that follows.

Inheritance from the product structure. For the cross-office channels $c \in \{M \cdot P, M \cdot G, M \cdot G \cdot P\}$ the shift Align_{mpt}^c is a product of tier-specific shifts. A sufficient primitive condition for (35)–(36) is that each component $\text{Align}_{mpt}^M, \text{Align}_{s(m),pt}^G, \text{Align}_{pt}^P$ separately satisfies the corresponding mean-independence condition and that the components are conditionally independent given q_k :

$$\mathbb{E}[\text{Align}_{mpt}^M \text{Align}_{pt}^P \mid \mathcal{H}_{m,t-1}, q_k] = \mathbb{E}[\text{Align}_{mpt}^M \mid \mathcal{H}_{m,t-1}, q_k] \cdot \mathbb{E}[\text{Align}_{pt}^P \mid q_k], \quad (37)$$

and analogously for Align^G and the triple product. Substantively this rules out strategic anticipation: for instance, the mayoral election outcome in m must not be systematically aligned with the federal incumbent's coat-tail strength beyond what q_k already absorbs. Including state $\times$ year fixed effects and pre-period vote shares in q_k is the natural way to discharge this condition.

Verticalização caveat for cycles 2002 and 2006. The product-independence assumption (37) is partially violated for the 2002 and 2006 electoral cycles by TSE’s *verticalização* rule (Resolution 20.993/2002, in force until EC 52/2006 took effect for the 2010 elections). During those cycles, state gubernatorial coalitions were required to mirror the national presidential coalition, so $\text{Align}_{s(m),pt}^G$ and Align_{pt}^P are not conditionally independent in the BJS sense: their joint distribution collapses toward perfect alignment by legal construction. The mechanical consequence is that $Z_{jmt}^{M\cdot G}$ and $Z_{jmt}^{M\cdot P}$ are more highly correlated for cycles 2002 and 2006 than for cycles 2010 and 2014, and the joint AR test on the four-channel cross-office stack pools *verticalizado* and *non-verticalizado* identifying variation. Two practical responses: (i) report the joint AR statistic with a footnote flagging the *verticalização* period; (ii) report a robustness AR test restricted to cycles 2010 and 2014 (post-*verticalização* sample), where (37) is constitutionally protected (EC 52/2006).

Within-cycle persistence (levels). In the levels specification for the pure mayoral channel, Align_{mpt}^M is non-zero for the entire ~ 4 -year mayoral cycle, so q_k must absorb anything that correlates persistently within a cycle with both alignment and outcomes (e.g. a slow-moving regional growth differential). Including a pre-period GDP trend, lagged $Y_{m,t-1}$, or muni-specific linear trends are natural strengthenings; the 2002-fixed baseline in $\omega_{fp,t}^c$ also reduces the dependence on contemporaneous alignment-cycle dynamics. For the cross-office channels, the inter-inauguration intervals are at most two years, so the within-interval persistence requirement is correspondingly weaker: the levels-versus-changes gap shrinks for $c \in \{M\cdot P, M\cdot G, M\cdot G\cdot P\}$.

Many-shocks count under Condition 2. The effective number of shocks differs sharply between the two variants. With $M = 5,570$, four mayoral cycles ($e_M \in \{2004, 2008, 2012, 2016\}$) and $P \approx 30$ parties, the total *candidate* shock count is on the order of $M \times P \times 4 \approx 6.7 \times 10^5$ for the pure mayoral channel. The levels variant uses each shock for ~ 4 post-election years, but for the pure mayoral channel those four year-replicates contribute redundant information for shock-level inference (the shift is constant within a cycle). The changes variant uses each shock once at inauguration, so its effective shock count is the same but its panel-level non-zero cell count is smaller by a factor of ~ 4 . For the cross-office channels the “constant within cycle” redundancy is already broken—the shift can flip mid-mayoral-term at the gubernatorial/presidential inauguration—so the levels-variant cell count is roughly $2\times$ rather than $4\times$ the changes-variant count, and the levels-versus-changes shock-level redundancy is correspondingly smaller. Condition 2 is therefore satisfied at comparable levels by both variants in terms of *distinct shocks*, but the changes variant has less redundancy and a more transparent shock-level interpretation. By contrast, the levels variant has more non-zero panel cells and thus more degrees of freedom for the muni-year regression—at the cost of a stronger Condition 1 requirement ((35)).

Operational rule. Levels instruments are the primary specification, on grounds of statistical power and continuity with the existing pipeline. The changes instruments are the robustness specification. If the two reject in agreement, the substantive conclusion is robust to whichever of (35) or (36) actually holds in the data; if they disagree, the discrepancy isolates exactly the part of the variation that (35) is asking about beyond (36).

2.7 Exclusion restriction

The exclusion restriction proper is that the instrument Z_{jmt}^ℓ enters Y_{mt} only through the sectoral employment composition s_{mt} , after conditioning on the volume control Vol_{mt} and on the controls of section 7. Three non-composition channels are conceivable and are addressed in turn:

1. *Volume of BNDES credit.* Political alignment changes both the composition and the level of BNDES credit. The level effect is partialled out by Vol_{mt} in (13). The exclusion restriction asks that, conditional on Vol_{mt} , Z_{jmt}^ℓ does not drive Y_{mt} through a residual volume channel.
2. *Non-BNDES political channels.* A new mayor may also change procurement, transfers, or local public investment in ways that move GDP. Condition 1 accommodates this only if those channels are orthogonal to the sector-specific pattern of $w_{jmp,t}^\ell$. We test this in robustness as a placebo: if Z_{jmt}^ℓ predicts municipal transfers or procurement after conditioning, the exclusion restriction is weakened.

3. *Direct partisanship effects on private behaviour.* Even absent any policy change, owners aligned with the new ruling party may invest more in their own firms because of confidence shocks. This would deliver a reduced-form effect on s_{mt} via firm-level investment that is not mediated by BNDES credit, but it is still a *composition* channel and therefore part of the estimand by construction—it is not an exclusion violation. The estimand is the GDP effect of compositional shifts driven by the political shock, whether transmitted through BNDES credit or through partisan-firm investment.

3 Instrument construction

This section assembles the sector-channel instruments (29)–(32) into the muni-year stack used by the AR test, and documents the cycle-alignment structure of the exposure weights.

Sector instruments and muni-level stacking. Equations (29)–(33) define the sector-level cross-office instruments. To take them to the muni-year level for the AR test, we *stack* sector and channel:

$$Z_{mt} \equiv (Z_{jmt}^c)_{c \in \mathcal{C}, j=1, \dots, J} \in \mathbb{R}^K, \quad K = |\mathcal{C}| \cdot J, \quad (38)$$

where $\mathcal{C} \subseteq \{\text{M}, \text{M} \cdot \text{P}, \text{M} \cdot \text{G}, \text{M} \cdot \text{G} \cdot \text{P}\}$ is the active channel set. For the currently leading post-D28 candidate margin `policy_block_active` $\times$ S3 with $J = 12$ active bins, $K = 48$ for the full four-channel cross-office stack ($\mathcal{C} = \{\text{M}, \text{M} \cdot \text{P}, \text{M} \cdot \text{G}, \text{M} \cdot \text{G} \cdot \text{P}\}$), $K = 36$ for the three-channel cross-office stack omitting the triple ($\mathcal{C} = \{\text{M}, \text{M} \cdot \text{P}, \text{M} \cdot \text{G}\}$), and $K = 12$ for the pure-mayoral specification ($\mathcal{C} = \{\text{M}\}$). At the secondary candidate margin `cnae_section` $\times$ S3 with $J = 51$ active bins, $K \in \{51, 153, 204\}$ for the three nested specifications.

Cycle alignment and exposure weights. Mayoral elections and gubernatorial/presidential elections sit on disjoint four-year cycles ($e_M \in \{2004, 2008, 2012, 2016\}$ versus $e_G = e_P \in \{2002, 2006, 2010, 2014\}$). Because governors and the president share the same election year, the pre-election windows T_t^G and T_t^P defined in (1) coincide and so $w_{jmt}^G = w_{jmt}^P$ when both are constructed on the common four-year window. Within any mayoral term, exactly one gubernatorial/presidential turnover occurs (at $t = e_M + 2$), so each cross-office shift has at most two within-mayoral-term changes: one at the mayoral inauguration $e_M + 1$ and one at the gubernatorial/presidential inauguration $e_M + 3$.

Remark 5 (Channel-set selection). The four-channel stack is the most aggressive specification and is the natural primary if the triple-cross-office shift $\text{Align}^{\text{M} \cdot \text{G} \cdot \text{P}}$ has enough non-zero panel cells to identify β . The three-channel stack drops the triple in case it is too thin (full-coalition alignment is the rarest configuration: it requires the firm-affiliated party to hold mayor, governor, and the federal coalition simultaneously). The pure-mayoral specification is the conservative comparison and continuity with the existing pipeline.

Remark 6 (Wide vs. aggregated muni-level instrument). An alternative aggregation, “stack-and-summarize,” would collapse Z_{jmt}^c to a single muni-year object via $Z_{mt}^c = \sum_j w_j Z_{jmt}^c$ for some weights w_j . This reduces K but imposes an aggregation that pre-loads which sectors matter; we treat it as a robustness check. The wide stacking (38) keeps the cross-sector pattern of the shock visible to the test and makes the “ $\beta = \mathbf{0}$ ” null intelligible coefficient by coefficient.

4 First stage

The first stage relating sectoral employment shares to the instruments is estimated on the muni-sector-year panel (Panel A) at the chosen aggregation margin:

$$s_{jmt} = \sum_{\ell} \pi_j^{\ell} Z_{jmt}^{\ell} + \sum_{\ell} \phi_j^{\ell} \text{EC}_{jmt}^{\ell} + \gamma_{jm} + \alpha_{jt} + \eta_{jmt}, \quad (39)$$

with municipality $\times$ sector fixed effects γ_{jm} (`~muni_id~cnae_section` in `fixest`), sector $\times$ year fixed effects α_{jt} (`~cnae_section~year`), and standard errors two-way clustered by `muni_id` and `sector`. The exposure control

$EC_{jm,t}^\ell$ absorbs the total pre-period political connectedness of (j, m) across all parties and is the level-restricted analogue of Goldsmith-Pinkham, Sorkin, and Swift’s share-level controls (Goldsmith-Pinkham, Sorkin, and Swift, 2020).

The first-stage F-statistic, computed via `fixest :: wald(., keep = "Z_")` on the full set of sector-tier instruments, is the diagnostic for [Condition 3](#) and for F2 informativeness in our identification chain. Per D20 it is not a validity gate for the AR test: weak first stages widen the AR confidence set but do not bias inference.

Remark 7 (Stacked first stage for the AR test). The first stage (39) need not be estimated separately to conduct the AR test: as the next section shows, it is internally embedded in the reduced-form Wald test. We document (39) here because the F-statistic from (39) is the relevant power diagnostic, and because the pattern of $\hat{\pi}_j^\ell$ across sectors is the natural mechanism narrative.

Why the cross-office form solves the FE-absorption problem. The standalone tier shifts Align^G and Align^P have weak cross-municipality variation that survives the standard FE structure $\gamma_{jm} + \alpha_{jt}$ in (39): Align_{pt}^P is constant across m at fixed t (a national indicator), and $\text{Align}_{s(m),pt}^G$ is constant within each (state, t) cell. In a tier-stacked instrument Z_{jmt}^P or Z_{jmt}^G , all surviving identifying variation must come from heterogeneity in the pre-period exposure shares $w_{jmp,t}^\ell$, which are themselves persistent and largely absorbed by γ_{jm} . The cross-office shifts in (14)–(16) inherit the full cross-municipality variation of Align^M and gate it on the higher-tier coalition state, producing instruments with rich cross-muni and cross-time variation that survive any plausible FE structure short of muni $\times$ year FE (which would absorb the endogenous variable itself). This is a mechanical reason—in addition to the substantive one above—to prefer the cross-office instruments to the additive tier stack.

5 The AR test: reduced form

5.1 Derivation

Step 1: Structural equation. At the muni-year level, restate (13) as

$$Y_{mt} = \alpha_m + \delta_t + \beta' s_{mt} + \lambda \text{Vol}_{mt} + \varepsilon_{mt}. \quad (40)$$

Stack the K sector-tier instruments as in (38). Aggregate the sector-level first stage (39) to the muni-year level by reading $s_{mt} \in \mathbb{R}^J$ as a vector and writing the muni-year first stage in matrix form:

$$s_{mt} = \alpha_m^{(s)} + \delta_t^{(s)} + \Pi Z_{mt} + \Phi EC_{mt} + \mu \text{Vol}_{mt} + \nu_{mt}, \quad (41)$$

where $\Pi \in \mathbb{R}^{J \times K}$ is the matrix of first-stage coefficients collected from (39) after the sector-by-sector slopes have been expanded into the wide format aligned with (38), and $EC_{mt} \equiv (EC_{jm,t}^\ell)_{\ell,j}$ is the stacked exposure-control vector.

Step 2: Reduced form by substitution. Substitute (41) into (40) and collect terms:

$$\begin{aligned} Y_{mt} &= \alpha_m + \delta_t + \beta' \left(\alpha_m^{(s)} + \delta_t^{(s)} + \Pi Z_{mt} + \Phi EC_{mt} + \mu \text{Vol}_{mt} + \nu_{mt} \right) + \lambda \text{Vol}_{mt} + \varepsilon_{mt} \\ &= \tilde{\alpha}_m + \tilde{\delta}_t + \underbrace{(\Pi' \beta)'}_{\equiv \gamma' \in \mathbb{R}^K} Z_{mt} + \underbrace{(\Phi' \beta)'}_{\equiv \psi'} EC_{mt} + \underbrace{(\beta' \mu + \lambda)}_{\equiv \tilde{\lambda}} \text{Vol}_{mt} + \underbrace{\beta' \nu_{mt} + \varepsilon_{mt}}_{\equiv \eta_{mt}}, \end{aligned} \quad (42)$$

where $\tilde{\alpha}_m = \alpha_m + \beta' \alpha_m^{(s)}$ and $\tilde{\delta}_t = \delta_t + \beta' \delta_t^{(s)}$ are absorbed by the muni and year fixed effects. Equation (42) is the *reduced-form GDP equation*.

Step 3: H_0 in the reduced form. Under $H_0 : \beta = \mathbf{0}$,

$$\gamma = \Pi' \beta = \mathbf{0} \in \mathbb{R}^K, \quad \psi = \Phi' \beta = \mathbf{0}, \quad \tilde{\lambda} = \lambda. \quad (43)$$

Equation (43) is the testable content. It says that under H_0 all K instruments enter the reduced-form GDP equation with zero coefficient. This is true *regardless of* Π : the implication holds whether the first stage is strong, weak, or zero.

Step 4: The AR test as the joint Wald on Z_{mt} . Estimate

$$Y_{mt} = \tilde{\alpha}_m + \tilde{\delta}_t + \gamma' Z_{mt} + \psi' \text{EC}_{mt} + \tilde{\lambda} \text{Vol}_{mt} + \eta_{mt}, \quad (44)$$

and test

$$H_0^{\text{AR}} : \gamma = \mathbf{0} \iff H_0 : \beta = \mathbf{0}. \quad (45)$$

The cluster-robust Wald statistic for $\gamma = \mathbf{0}$ in (44) is the AR test statistic. Operationally:

```
mod_rf <- feols(log_gdp_pc ~ Z + EC + Vol | muni_id + year,
               data = dt, vcov = ~muni_id)
ar_stat <- fixest::wald(mod_rf, keep = "^Z_")$stat
```

5.2 Equivalence with the canonical AR statistic

The canonical Anderson–Rubin statistic of [Anderson and Rubin \(1949\)](#) for $H_0 : \beta = \beta_0$ in the simultaneous equation $Y = X\beta + u$ with instruments Z is

$$\text{AR}(\beta_0) = \frac{(Y - X\beta_0)' P_Z (Y - X\beta_0) / K}{(Y - X\beta_0)' M_Z (Y - X\beta_0) / (N - K)}, \quad (46)$$

with $P_Z = Z(Z'Z)^{-1}Z'$ and $M_Z = I - P_Z$ after projection of all variables on the fixed-effect space. Under correct specification and $\beta = \beta_0$, $\text{AR}(\beta_0) \sim F_{K, N-K-n_{\text{FE}}}$ exactly under homoskedasticity, and asymptotically χ_K^2 / K under heteroskedasticity-robust forms ([Andrews, Stock, and Sun, 2019](#); [Kleibergen, 2002](#)).

Lemma 1 (Reduced-form equivalence at $\beta_0 = \mathbf{0}$). *At $\beta_0 = \mathbf{0}$, the AR statistic (46) reduces to the F -statistic for $\gamma = \mathbf{0}$ in the reduced-form regression (44).*

Sketch. At $\beta_0 = \mathbf{0}$, $Y - X\beta_0 = Y$ and (46) becomes the ratio of the explained sum of squares of Y on Z (divided by K) to the residual sum of squares (divided by $N - K$), which is exactly the F -statistic for the joint significance of Z in $Y = \gamma'Z + (\text{FE}) + \eta$. With heteroskedasticity-robust / cluster-robust variance, the corresponding Wald form replaces the homoskedastic F and is the cluster-robust AR statistic of [Finlay and Magnusson \(2009\)](#). $\square$

What this buys us. [Lemma 1](#) is what makes the AR optimality test cheap to implement: the practitioner only needs to estimate (44) and run a Wald test on the instruments. No first stage is needed for the test; no $\hat{\beta}$ needs to exist. The first stage (39)–(41) is informative for power but not for size.

5.3 Weak-instrument robustness, many instruments, and CLR

The AR test has correct size under $H_0 : \beta = \mathbf{0}$ for any value of Π , including $\Pi = 0$. This follows from the fact that under H_0 the reduced-form coefficient $\gamma = \Pi'\beta$ vanishes regardless of Π . The 2SLS Wald test is not size-correct when Π is small, because its critical values rely on a non-degenerate first-stage limit ([Andrews, Stock, and Sun, 2019](#); [Dufour, 1997](#)).

For $K > 1$, the conditional likelihood ratio (CLR) test of [Moreira \(2003\)](#) dominates AR in power for many parameter configurations. We treat CLR as a complementary robustness statistic; it is not yet implemented in the project pipeline.

For $K \rightarrow \infty$ asymptotics (the secondary margin `cnae_section` $\times$ S3 with $K \approx 50$ –150), [Mikusheva and Sun \(2022\)](#) provide the ridge-regularised jackknifed AR test that is valid for $K/N \rightarrow c \in (0, 1)$. This is the relevant inference for the many-instrument robustness specification.

6 Volume control

The volume term Vol_{mt} in (13) partitions the political shock’s effect on Y_{mt} into a level (or volume) channel and a composition channel. Without it, the test of $H_0 : \beta = \mathbf{0}$ would conflate the two: turnover both reallocates credit across sectors and changes its aggregate amount, and (5) concerns only the former.

Construction.

$$\text{Vol}_{mt} \equiv \frac{\sum_j \sum_{f \in \mathcal{F}(j,m)} \text{BNDES}_{f_{mt}}}{\text{GDP}_{m,2002}} = \frac{\text{BNDES_total}_{mt}}{\text{GDP}_{m,2002}}. \quad (47)$$

The numerator is current-period total municipal BNDES disbursements; the denominator is initial-period municipal GDP (2002 R\$). The ratio is unit-free and time-comparable across municipalities. The choice of the 2002 denominator fixes the scaling at a pre-treatment value, so Vol_{mt} varies in t only through the numerator.

Remark 8 (Why a pre-treatment denominator). Using contemporaneous GDP_{mt} as the denominator would induce $\text{cor}(\text{Vol}_{mt}, Y_{mt}) \neq 0$ mechanically through the denominator, bias the partialling-out of the volume channel, and create a “bad-control” problem for the AR test. The 2002 denominator avoids this.

Recipient-class composition of the numerator (D5-op). The Phase 0 BNDES recipient audit (A0.4) decomposes total disbursements over 2002–2017 into productive-firm (71.6%), public-entity (28.3%), financial-institution intermediation (0.10%), and residual ($< 0.01\%$) classes. The financial-intermediation overlap check confirms no double counting above 0.018% of total credit in any year. Under D5-op (user, 2026-05-12) the *primary* volume control aggregates all classes:

$$\text{Vol}_{mt}^{\text{joint}} \equiv \frac{\text{BNDES_total}_{mt}}{\text{GDP}_{m,2002}}, \quad (48)$$

with the productive-firm component restricted to RAIS-merged disbursements to remain internally consistent with the D1 RAIS-only analyzed universe. The *split-volume* robustness specification enters three terms separately: (i) *productive-nonRAIS*, private + CNAE recipients without a RAIS row in year t ($\sim 42\%$ of D3.1’s productive aggregate); (ii) *financial-intermediary*; (iii) *public-entity*. Public-credit flows outside the BNDES envelope (CEF, BB, BNB, constitutional funds) are *not* in Vol_{mt} ; they are registered as out-of-scope in the project blueprint (open angle on alternative public-credit channels) and re-enter the discussion under [subsection 9.2](#).

Should Vol_{mt} be instrumented? Vol_{mt} is itself endogenous: the same political shocks that move sector shares move the aggregate. The natural instrument for Vol_{mt} is a muni-level aggregation of Z_{jmt}^ℓ , weighted by the pre-period sector shares; this is the muni-level shift-share for the level channel. The four candidate operationalisations from the strategy memo (A10) are:

1. *Pure AR*. OLS for both s_{mt} and Vol_{mt} in the reduced form (i.e., neither is treated as the structural endogenous variable). The AR test is on Z_{mt} ; Vol_{mt} is included as a pre-determined control.
2. *Partial IV (baseline, per D24)*. Treat s_{mt} as endogenous and instrumented by Z_{mt} ; treat Vol_{mt} as exogenous. The reduced-form AR equation (44) includes Vol_{mt} on the right-hand side and tests $\gamma = \mathbf{0}$ on Z_{mt} .
3. *Full IV*. Instrument both s_{mt} and Vol_{mt} . This requires a separate instrument for Vol_{mt} that is not collinear with Z_{mt} in finite sample; the obvious candidate is a sum or PCA of Z_{jmt}^ℓ across j , but that creates near-collinearity with Z_{mt} itself.
4. *Mixed*. OLS on s_{mt} , IV on Vol_{mt} . Symmetric to (3) and similarly suffers from near-collinearity.

The baseline (2) is the operative specification of the paper. Approaches (1), (3), (4) enter as robustness. The choice between (1) and (2) is consequential only if Vol_{mt} is a strong correlate of ε_{mt} *after* partialling out Z_{mt} and the FE; under the exclusion restriction this residual correlation is bounded by the strength of non-political confounders of total credit.

Remark 9 (The collinearity tradeoff). Approaches (3) and (4) deliver consistent estimators of both β and λ if a separate instrument for Vol_{mt} exists. The candidate $\sum_j Z_{jmt}^\ell$ collapses across sectors and discards the identifying variation in cross-sector tilts that is precisely what powers the AR test. Building a non-trivial second instrument requires new variation (e.g. a national fiscal-cycle shock that scales the BNDES envelope without shifting its sectoral allocation). We do not pursue this here.

7 Controls and fixed effects

We document the role of each control / fixed effect and the assumption it enables.

Municipality fixed effect α_m . Absorbs all time-invariant municipal characteristics: geography, fixed infrastructure stocks, time-invariant institutional quality, fixed sectoral-comparative-advantage components of θ_m . With α_m , identification of β uses only *within*-muni variation in s_{mt} . Required for [Condition 1](#) to be plausible: a level correlation between alignment frequency and economic level across municipalities is allowed.

Year fixed effect δ_t . Absorbs national-level shocks affecting all municipalities in the same year: common monetary policy, national IPCA inflation, national electoral cycles affecting all jurisdictions simultaneously. Required so the identifying variation in Align_{mpt}^ℓ is the cross-muni dispersion in who-rules-where in a given year, not the national alternation between Lula, Dilma, and Temer.

Exposure control $\text{EC}_{jm,t}^\ell$. $\text{EC}_{jm,t}^\ell = \sum_p w_{jmp,t}^\ell$ measures total pre-period political connectedness of (j, m) at office ℓ . It is the level-restricted analogue of [Goldsmith-Pinkham, Sorkin, and Swift](#)'s share-controls strategy: it absorbs any direct effect of the share component of the SSIV that is not mediated by the shift component. In the AR reduced form (44), $\text{EC}_{jm,t}^\ell$ is included alongside Z_{jmt}^ℓ to remove this share-channel confounding from the test on γ .

Pre-period GDP trend (robustness). A muni-specific linear trend in t , fitted on $Y_{m,t-1}$ and earlier, can be included to harden [Condition 1](#) against persistent muni-level growth differentials that correlate with party alignment. This is the practical manifestation of the q_k in [Condition 1](#). We treat it as robustness because it costs degrees of freedom (1 parameter per municipality) and because the muni-by-year FE design already absorbs everything cross-cycle.

State $\times$ year fixed effect (robustness). $\zeta_{s(m)t}$ where $s(m)$ is the state of m absorbs governor-level shocks common to all municipalities in a state. Including it removes the variation exploited by the governor-tier instruments Z_{jmt}^G , so it is incompatible with using all three tiers; it is included as a within-state robustness exercise that uses only mayor-tier instruments.

What is not included by design. We do not include lagged sector shares $s_{m,t-1}$ as controls. Doing so would absorb the pre-period mix and break the within-simplex projection: the gradient $\nabla_s Y$ is identified from changes *around* the pre-period mix, not from levels conditional on it. This is the standard “Nickell bias”-style argument adapted to the simplex.

8 Inference

Primary clustering. Standard errors in (44) are clustered by `muni_id`. The number of clusters ($M = 5,570$) is large relative to the parameter dimension and to the number of effective shocks per cycle, so cluster-robust inference is well-calibrated. [Finlay and Magnusson \(2009\)](#) establish that the cluster-robust Wald statistic for $\gamma = \mathbf{0}$ in (44) is the cluster-robust AR test, with size correctly controlled under quite general within-cluster dependence structures.

Two-way clustering. For the sector-level first stage (39) we cluster two-way by municipality and sector. For the muni-year reduced form (44) the natural second clustering dimension is harder: clustering by *shock* (ℓ, p, m, t) is collinear with the by-muni clustering, and clustering by year would yield only $T = 16$ clusters.

Spatial correlation. Adão, Kolesár, and Morales (2019) show that residuals in shift-share designs are correlated across units that share similar exposure structures (here, similar $w_{jmt,t}^\ell$ profiles). Cluster-robust SEs by `muni_id` do not fully internalize this correlation, and may understate uncertainty. Two remedies:

1. State-level clustering as a robustness check. The 27 Brazilian states are coarse and trigger few-clusters concerns (Carter, Schnepel, and Steigerwald, 2017); we use the wild cluster bootstrap in this case.
2. The Adão–Kolesár–Morales correction (Adão, Kolesár, and Morales, 2019), computed at the shock level. This is the right inference under the BJS framework but is not yet implemented in the pipeline.

Concern: state-level correlation and the cross-office channels. The standalone tier instruments Z_{jmt}^G and Z_{jmt}^P have weak cross-municipality variation that survives the standard FE structure— Align_{pt}^P is constant across municipalities at any t and so is absorbed by δ_t , and $\text{Align}_{s(m),pt}^G$ is constant within each (state, t) cell—which is the mechanical reason we drop them from the primary stack in favor of the cross-office shifts of subsection 2.1. The cross-office instruments Z_{jmt}^{M-P} , Z_{jmt}^{M-G} , and Z_{jmt}^{M-G-P} inherit the cross-municipality variation of Align_{mpt}^M and gate it on the higher-tier coalition state, so they survive any plausible FE structure short of `muni`×`year` FE. The Adão–Kolesár–Morales correction (Adão, Kolesár, and Morales, 2019) remains the right inference under the BJS framework when within-state residual correlation is a concern; it is applied at the shock level, where shocks now correspond to channel-party-municipality-time cells $k = (c, p, m, t)$. We report *channel-specific* AR statistics for each $c \in \mathcal{C}$ alongside the joint test, with the pure-mayoral statistic as the conservative benchmark and the cross-office statistics as the mechanism-relevant hypotheses.

Effective sample size and time-series content. The levels instrument Z_{jmt}^c is a step function in t that is piecewise constant between consecutive elements of E_c . With `muni` and `year` FE, the variation exploited is the cross-municipality dispersion in instrument step-changes at the inauguration boundaries: for the pure mayoral channel, mayoral inaugurations {2005, 2009, 2013, 2017}; for the cross-office channels, the union of mayoral and gubernatorial/presidential inaugurations {2003, 2005, 2007, ..., 2017}. Each municipality contributes ~ 3 –4 independent step-changes for the pure mayoral channel and ~ 7 –8 for each cross-office channel; the cross-municipality average has $\sim M \times 3$ degrees of freedom for the mayoral channel and $\sim M \times 7$ for each cross-office channel. This is the relevant denominator for power, not the panel-level $M \times T$.

9 Robustness and limitations

9.1 Robustness exhibits

The estimation pipeline produces four robustness blocks that the AR test is evaluated against. Each is documented operationally; the specific F-statistics and rejection probabilities below are drawn from the Phase 1 exploration (`explorations/anderson_rubin/active_denominator/`) at the production margin `policy_block_active` × S3 and the side-by-side `cnae_section` × S3.

R1. Denominator variants. The three skeleton variants of Equation 2 (contemporaneous, frozen, balanced) are estimated as separate AR tests on the same instrument stack. Phase 1 finds that all three variants reject the optimality null at conventional levels, with the headline AR F -statistic on the `muni` panel at ≈ 2.69 for the `muni_id`×`cnae_section` FE-and-clustering configuration. The contemporaneous variant is the primary; the frozen and balanced variants are reported alongside, since their first stages drop the post-election attrition from balanced-panel restriction ($\sim 52\%$ at cycle 2013).

R2. Margin choice: policy block vs. CNAE section. The production margin uses 12 firm-side policy blocks (`policy_block_active` × S3, with one block held out as the simplex base j_0), so the active instrument set has $K_{\text{eff}} = 9$ blocks × at most four cross-office channels. The side-by-side robustness margin is `cnae_section` × S3 ($J = 51$ active bins). Production AR F -statistics are 4.37 ($p = 2 \cdot 10^{-4}$) at policy block and 2.05 ($p = 2.1 \cdot 10^{-4}$)

at CNAE section. The Phase 1 standalone diagnostic at policy block (C2.1.5) reports $F = 4.19$; drop-top-1-shock $F = 3.37$ ($p < 0.001$, reject); drop-top-2 $F = 2.72$ ($p < 0.01$, reject). Toggling the BHJ §4.4 cell-level slack control on and off shifts F by $\Delta F = 0.023$ at the production FE configuration, within the Phase 1 envelope.

R3. Joint vs. split volume control. The default specification enters $\text{Vol}_{mt}^{\text{joint}}$ from (48); the robustness specification enters productive-nonRAIS, financial-intermediary, and public-entity disbursements separately, each normalized by initial-period GDP. Rejection regions are qualitatively stable at both aggregation margins: joint and split AR tests both deliver $p \approx 2 \cdot 10^{-4}$. The financial-intermediary channel coefficient in the split-volume reduced form is -0.018 (significant at the 5% level), qualitatively matching the Phase 1 OLS exploration’s -0.105 on the same channel.

R4. Pre-trend characterization (documentation, not a gate). Per the user’s 2026-05-12 instruction, pre-trend evidence is documented rather than used as an admissibility gate. Phase 1 reports: the governor pre-trend rejection initially flagged in B17 resolves as a specification artifact under strict-timing windows; the presidential pre-trend is genuine but moderate ($p < 0.005$ on $\Delta \log \text{GDP}$); the mayoral pre-trend is clean ($p = 0.80$). As a stress test, dropping the high-pre-trend presidential CNAE sections delivers a first-stage $F = 53.84$ ($p < 10^{-7}$) on the AR-robust specification. Substantively, the AR identification condition is conditional mean-independence of the shock (35) or (36) rather than zero pre-trends, so the presidential moderate pre-trend is flagged in [subsection 9.2](#) rather than treated as a failure.

9.2 Limitations

L1. RAIS-formal-only bound. The endogenous variable s_{jmt}^{emp} is constructed from the formal-sector RAIS panel. Politically driven shifts in informal employment, MEI-unfiled activity, and RAIS-Negativa-only firms are outside the analyzed channel. The upper bound on the Negativa-recoverable firm mass is 7.64% of admin-observed firm-years (A0.1) and the corresponding employment-mass impact is 1.83% (A0.5). The bound is small enough that informal-sector mismeasurement is unlikely to overturn the rejection at the contemporaneous margin, but the structural claim is in any case narrower than “the sectoral mix is GDP-optimal”—it is “the formal-sector composition channel is GDP-neutral conditional on volume”.

L2. Other public-credit channels. The volume control Vol_{mt} covers BNDES disbursements only. Parallel public-credit envelopes—Caixa Econômica Federal (CEF), Banco do Brasil (BB), Banco do Nordeste (BNB), and the constitutional regional funds (FCO, FNE, FNO)—are not in Vol_{mt} and are not in the instrument. If the political shock drives these channels with the same cross-sector tilt as Z_{jmt}^c generates for BNDES, the AR test attributes their composition effect to the BNDES channel. This is a known limitation of the formal-channel estimand and is registered as a future-scope open angle in the project blueprint.

L3. Pre-trend residual on the presidential channel. [Proposition 1](#) states the formal sufficient condition for AR identification as conditional mean-independence of the shock from prior trajectories given q_k . A non-zero pre-trend on the presidential channel indicates that some component of Align_{pt}^p is forecastable from pre-period dynamics absorbed by q_k . With state $\times$ year fixed effects in q_k , the residual presidential pre-trend is moderate ($p < 0.005$) and the robustness drop-section exercise (R4) preserves the rejection. The presidential-channel coefficients are nonetheless to be read with this qualification, and the cross-office channels M·P and M·G·P inherit the same qualification through the product structure (37).

L4. Cluster-robust SEs vs. AKM shock-level inference. The primary specification reports SEs clustered by municipality ([section 8](#)). At $K_{\text{eff}} = 9$ policy-block instruments, the standard two-way muni + year clustering widens AR t -statistics by approximately 50% relative to muni-only clustering, but the headline 5% rejection survives. The [Adão, Kolesár, and Morales \(2019\)](#) shock-level correction, implemented via the `ssaggregate` routine,

is the right inference under the BJS framework and is flagged for the published version; it is not in the current pipeline.

L5. First-stage strength on the emp-share endogenous variable. The Phase 1 first stage at script 53 returns an effective F in the interval $[0.03, 3.38]$ for s_{jmt}^{emp} on the policy-block stack. This is below the conventional strong-instrument threshold. The AR test is robust to weak instruments by construction (Andrews, Stock, and Sun, 2019)—it has correct size for any value of Π , including $\Pi = 0$ —so weak-instrument inference is the right frame for the headline result. We disclose the first-stage F alongside the AR statistic and do not report a 2SLS point estimate at this margin.

10 Summary

The Anderson–Rubin optimality test of this paper is operationally a Wald test on the instruments in the reduced-form GDP regression (44). The substantive content is the directional-derivative condition (5) of section 1, which equates the local-optimality null to $\beta = \mathbf{0}$ in the structural equation (13). The reduced-form equivalence (Lemma 1) is what makes the test (i) weak-instrument robust, (ii) joint across K sector-tier instruments, and (iii) implementable without producing a $\hat{\beta}$ point estimate when the first stage is weak. The shift-share framework of Borusyak, Hull, and Jaravel (2022) provides the identification conditions; Proposition 1 states the formal sufficient conditions for the levels and changes timing variants and shows that the changes variant carries a milder requirement on pre-period dynamics. The volume control Vol_{mt} partials out the level channel of the political shock, leaving only the composition channel as the object of the test. Inference is cluster-robust by municipality, with state-level and Adão, Kolesár, and Morales (2019) corrections in robustness.

Appendix: Notation glossary

Symbol	Object
m, j, t, ℓ, p	municipality, sector, year, office, party
f	firm
$\mathcal{F}(j, m), \mathcal{F}_{jm,e}^{\text{pre}}$	firm support of (j, m) / pre-period support at inauguration e
s_{jmt}	sector j employment share in (m, t) , $\sum_j s_{jmt} = 1$
s_{jmt}^B	sector j BNDES credit share in (m, t)
Y_{mt}	log real GDP in (m, t)
β	J -vector of GDP gradient w.r.t. employment shares
Vol_{mt}	$\text{BNDES_total}_{mt} / \text{GDP}_{m,2002}$
T_t^ℓ	pre-election window for office ℓ at year t , eq.(1)
$L_{f,p,t}, L_{f,t}$	owners of firm f affiliated with party p / total owners at t
$\omega_{fp,t}^\ell$	firm-level baseline party exposure, pooled-count, eq.(17)
$N_{jm,t}^\ell$	owner-count normalizer in cell (j, m) , eq.(22)
$w_{jmp,t}^\ell$	sector-level baseline party exposure, eq.(21)
FA_{fmt}^c	firm-level cross-office instrument for channel c , eq.(20)
Align_{mpt}^ℓ	$\mathbf{1}\{\text{party } p \text{ holds office } \ell \text{ in } m \text{ at } t\}$
$\Delta \text{Align}_{mpt}^\ell$	alignment turnover ($\neq 0$ at inauguration)
Align_{mpt}^c	cross-office shift for channel c , eqs.(14)–(16)
Z_{jmt}^c	sector-channel levels instrument, eqs.(29)–(32)
ΔZ_{jmt}^c	sector-channel changes instrument, eq.(33)
Z_{mt}	stacked muni-year instrument vector, eq.(38), dim. K
$\text{EC}_{jm,t}^\ell$	sector-level exposure control, $\sum_p w_{jmp,t}^\ell$
Π	first-stage matrix in eq.(41)
$\gamma = \Pi' \beta$	reduced-form coefficient on Z_{mt} in eq.(44)
AR, F_{AR}	Anderson–Rubin statistic / F -form
$H_0 : \beta = \mathbf{0}$	local-optimality null

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